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Loop 1: Repetition Structure

Programming Logic · Week 8 · 4 November 2026

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Learning Outcomes

Week 8 · Programming Logic

By the end of this session, you can make a program repeat steps without copying code.

Explain

Why repetition matters and how a loop runs

Write

WHILE and FOR loops in pseudocode

Design

Flowcharts and trace tables for counting and summing

```
count = 1
WHILE count <= 5 DO
    OUTPUT count
    count = count + 1
END WHILE
```

Anatomy of a Loop

STEP 01

Initialize

Set the starting value before the loop begins.

STEP 02

Check

Test the condition. If true, run the body again.

STEP 03

Update

Change the variable so the loop can end.

```
total = 0
count = 1
WHILE count <= 5 DO
    total = total + count
    count = count + 1
END WHILE
OUTPUT total
```

WHILE Loop

WHILE checks the condition before every round. If it is false at the start, the body never runs.

```
total = 0
FOR i = 1 TO 5 DO
    total = total + i
END FOR
OUTPUT total
```

FOR Loop

FOR puts the start, end and step in one line. Use it when the number of rounds is known in advance.

Tracing a Loop

Counting

count	Output
1	1
2	2
3	3
4	4
5	5
6	stop

Summing

i	total
1	1
2	3
3	6
4	10
5	15
6	stop

Condition

i	i <= 5
1	True
2	True
3	True
4	True
5	True
6	False

Common Loop Mistakes

Mistake 1

Infinite loop

The condition never becomes false, so the loop never stops.

Mistake 2

No update

Forgetting count = count + 1 is the most common cause.

Mistake 3

Off-by-one

Using < instead of <= skips the last round.

Mistake 4

Bad start

A wrong initial value shifts every result.

Mistake 5

Sum not zero

Set total = 0 before the loop, or old values leak in.

Mistake 6

Wrong loop

FOR for a known count, WHILE when a condition decides.

Number Series Problem

Input a number N, then use loops to produce a number series.

Tasks

- Display all numbers from 1 to N
- Display the sum of 1 to N
- Display only the even numbers from 1 to N

Deliverables

- Pseudocode for all three tasks
- Trace table for N = 5


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